

COMPUTER NETWORKS

26 & 27-Feb-2026

Address Classes

- Class A - large organizations , governments
- Class B - medium sized organizations
- Class C - small organizations

Total number of hosts & Networks

- 1 address is the **network address**
- 1 address is the **broadcast address**

Number of hosts = $2^{\text{number of host bits}} - 2$

Number of networks = $2^{\text{variable no. Of bits for network}}$

Total Class B networks = 6 + 8

$2^{14} = 16,384$ networks in Class B

IPv4 networks

A → 7 bits

B → 14 bits

C → 21 bits

Number of networks = $2^{\text{variable bits}}$

IPv4 = 32 bits total

Total networks:

Class A	2^7	128
Class B	2^{14}	16,384
Class C	2^{21}	2,097,152

Class A → few networks, many hosts

Class B → medium networks, medium hosts

Class C → many networks, few hosts

Default Subnet Mask

- This is how a computer knows which part is the network and which part is the device
- Before subnetting was invented, each class had a built-in mask
- This built-in mask is called the **default subnet mask**
- Mask bit **1** → Network
- Mask bit **0** → Host

Default Subnet Mask/CIDR notation

- Classless Inter-Domain Routing

Class	Default Mask	CIDR	Network Bits
Class A	255.0.0.0	/8	8
Class B	255.255.0.0	/16	16
Class C	255.255.255.0	/24	24

Default Subnet Mask

If IP starts with...	Class	Default Mask
1–126	A	255.0.0.0
128–191	B	255.255.0.0
192–223	C	255.255.255.0

Example

- What is the default subnet mask of 172.20.5.6?
- Class?

Example

- What is the default subnet mask of 172.20.5.6?
- 172 → Class B
- Default mask ?

Example

- What is the default subnet mask of 172.20.5.6?
- 172 → Class B
- Default mask → 255.255.0.0

Calculating Network Address

- The **network address** is the first address in a subnet
- It identifies the network itself (not a device)
- You calculate it using:
- IP address AND Subnet mask

Example-1

IP: 192.168.10.25

Mask: 255.255.255.0 | (/24)

Step 1: Write down the Mask

255.255.255.0 means:

- First 3 octets = Network
- Last octet = Host

Step 2: Set Host Part to 0

192.168.10.0

Alternate way

11000000 . 10101000 . 00001010 . 00000000 (IP)

AND

11111111 . 11111111 . 11111111 . 00000000 (Mask)

11000000 . 10101000 . 00001010 . 00000000

Decimal notation:

192 . 168 . 10 . 0

Network Address = 192.168.10.0

Class A

- IP Address : 10.2.100.17
- Default Class A subnet mask : 255.0.0.0
- DBA : 10.255.255.255

Class B

- Class B can be diagnosed from first decimal value:
- 128 - 191 → 64 values in the first octet

10 000000 = 128

10 000001 = 129

10 000010 = 130

10 000011 = 131

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10 111111 = 191

Class B

- For a network starting with 128
- Networks will be : 128.0.0.0 - 128.0.255.255
- first IP address denotes the network itself
- last IP address is broadcast address
- remaining are $65536 - 2 = 65534$ hosts (usable in this network)

Class B

- 130.1.2.3 (given host IP address)
- class B from 130
- To find out the network address
- Default mask = 255.255.0.0 (from Class B)
- 130.1.2.3
- AND
- 255.255.0.0
-
- 130.1.0.0 (bcz AND with zero)
- **this is network address**
- so 130.1.2.3 belongs to 130.1.0.0 network
- if this address is for BNU then whole world knows you as **130.1.0.0**
- 130.1.0.0 ----- 130.1.255.255 : Possible IP addresses in this network
- Network Address → 130.1.0.0
- Directed broadcast Address → 130.1.255.255

Class C

- 192.168.10.11
- Network Address → 192.168.10.0
- default Class C subnet mask 255.255. 255.0
- DBA = 192.168.10.255

Remember for IPv4

- If all the bits in the host part are "0", that represents the network address
- If all the bits in the host part are "0" except the last bit, it is the first usable IP address
- If all the bits in the host part are "1" except the last bit, it is the last usable IP address
- If all the bits in the host part are "1", that represents the DBA

Class Exercise

- Write an IP address that belongs to class B
- Write the default Subnet Mask
- Write the Network Address

Sample solution

TASK	SOLUTION-Sample
Write an IP Address that belongs to Class-B	172.16.189.5 (128-191 → Class B)
Write the default Subnet Mask	255.255.0.0
Write the Network Address	172.16.0.0 (Last 2 octets in above address → zero)

Examples

Q1

IP: 192.168.5.23

Mask: 255.255.255.0

Network Address: _____

Q2

IP: 10.45.78.9

Mask: 255.0.0.0

Network Address: _____

Q3

IP: 172.16.200.15

Mask: 255.255.0.0

Network Address: _____

Q4

IP: 192.168.1.199

Mask: 255.255.255.0

Network Address: _____

Answers

Q1: 192.168.5.0

Q2: 10.0.0.0

Q3: 172.16.0.0

Q4: 192.168.1.0

Book-B

- Chapter 21
- Douglas E. Comer